

CARVIS

Clinical Anatomical Rendering & Visualization Intelligent System

Çankaya University · Department of Computer Engineering · 2025–2026



Team

Alperen Berke Çetinkaya · Muhammed Yusuf Özcan
· Sezer Ataş · Mete Serpil

Advisor

Dr. Gül Tokdemir

AI in Medicine Has a Numbers Problem

The Scenario

A language model says "**tumor volume is 47 cm³.**" The sentence sounds confident. The number might be wrong. You cannot tell, because the model speaks the same way regardless.

For a radiologist, a wrong number changes the clinical picture and the decision that follows.

Why Hospitals Avoid These Tools

Hospitals are not avoiding these tools because they know too little. They avoid them because the tools are **unreliable**.

- ⊗ Unreliability in medicine is more dangerous than incomplete knowledge. That is the problem this project exists to solve.

Two Paths, Both Flawed

The existing literature forces a binary choice. Neither option is acceptable for clinical deployment.

End-to-End LLM

Fluent and flexible but carries **high hallucination risk**. Output cannot be verified. The model speaks with equal confidence whether it is right or wrong.

Rigid Template

Safe with **zero hallucination** but mechanical. Cannot answer questions or adapt to individual cases. Honest, but silent.

The field asks you to choose between **fluent but dishonest** and **honest but silent**. We did not choose either.

A Number Without Context Is Not a Measurement

A segmentation model outputs "**47 cm³.**" No error range. No source. No validation. The number looks identical whether it was computed correctly or hallucinated entirely. A clinician cannot act on a number alone.

Four Things Required to Make a Measurement Clinically Usable

1

How much should I trust this?

What the AI gives you : the raw output.

2

Where did it come from?

So you know how much to trust it.

3

If it's wrong, how would I know?

So you can verify where it came from.

4

What if I ask you something you shouldn't answer?

So the system does not guess when the question is outside its scope.

Separate What the Model Does From What the Code Does

The Core Idea

Numbers come from code. Language comes from the model. A validation layer sits between them.

✔ The LLM is an enhancement, not a dependency.

Compute First

Every measurement is computed before the model sees anything.

Narrate Second

The model writes narrative around a structured facts object.

Validate Always

Wrong numbers are corrected.
If output cannot be repaired, the system falls back to a code-generated report.

CARVIS SAMPLE REPORT & CLINICAL QA

CARVIS Clinical AI

Case: BraTS20_Validation_001 May 22, 2026

Clinical Context

This mr brain tumor review is based on FLAIR, T1, T1ce, T2. The study date is unavailable in the current metadata. A clinical note is available for contextual correlation. No prior comparison material accompanies the current package.

PATIENT DEMOGRAPHICS & SCAN METADATA

- **Patient ID:** BraTS20_Validation_001
- **Study ID:** BraTS20_Validation_001
- **Study Date:** Unknown
- **Modality:** MR (FLAIR, T1, T1ce, T2 sequences)
- **Available imaging inputs:** 0 DICOM series directories and 4 NIFTI files.
- **Clinical note present:** yes; prior reports present: no.
- **Report detail mode:** balanced.
- **Purpose:** case-level brain tumor imaging assessment with automated segmentation review.

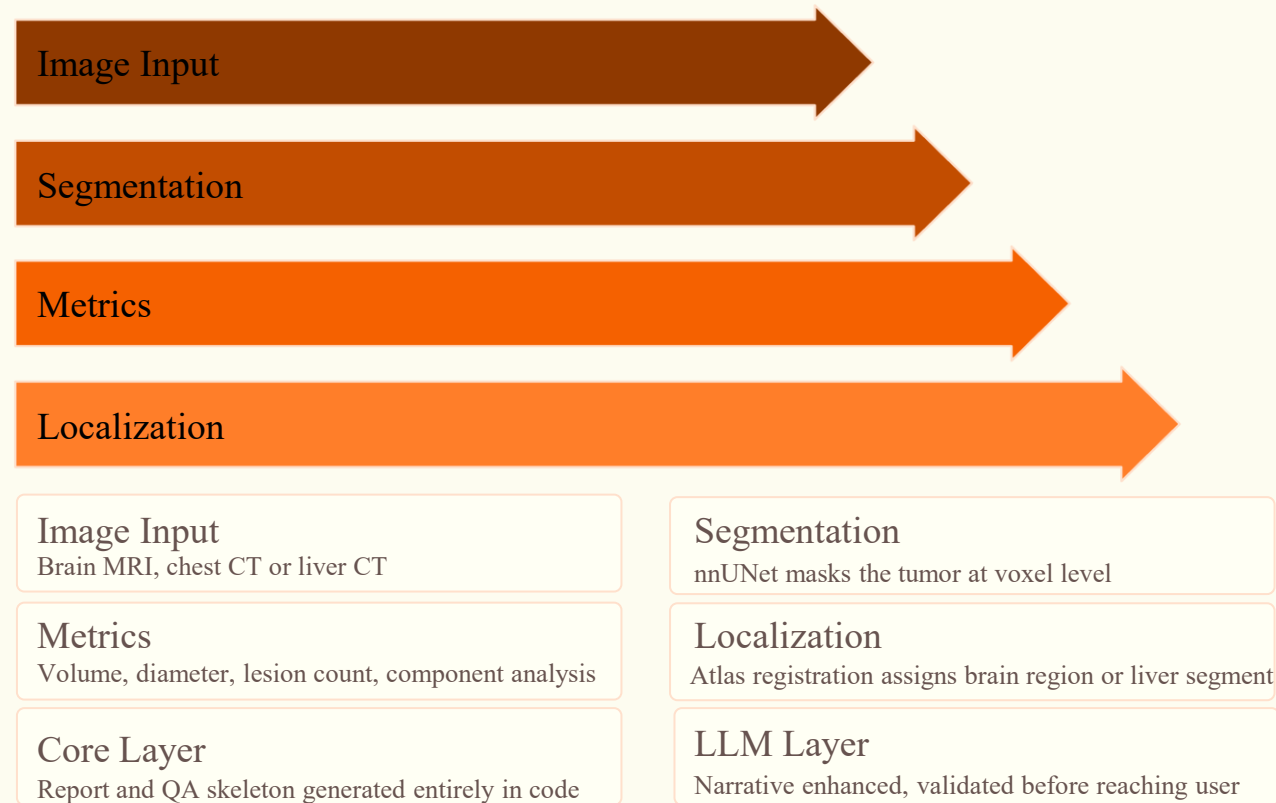
Findings

The dominant pattern localizes to Left Frontal lobe (Left Frontal Pole), subcortical, middle level. The segmented appearance is best described as multiple lesion groups with fragmented internal segmentation components. The qualitative burden pattern is predominantly non-enhancing/infiltrative with substantial edema burden. Approximate regional overlap is concentrated in Left Frontal Pole, Left Cerebral White Matter, Left Middle Frontal Gyrus. Any mass effect or midline shift

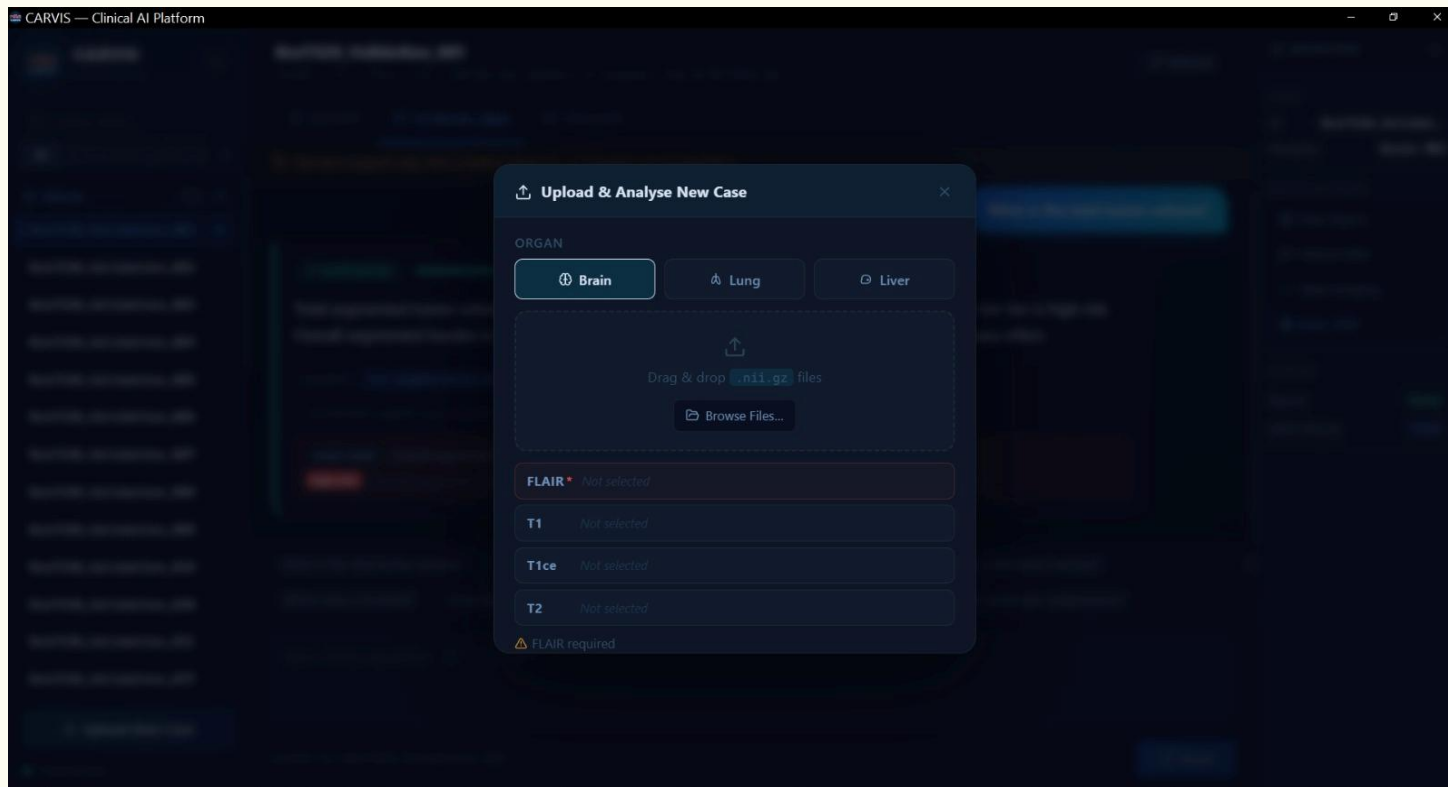
The screenshot displays the CARVIS Clinical AI Platform interface. The top header shows the case name "BraTS20_Validation_001" and a "Refresh" button. Below the header, there are tabs for "REPORT", "CLINICAL Q&A", and "IMAGING". The "CLINICAL Q&A" tab is active, showing a decision-support only message: "Decision-support only. Not a medical diagnosis or treatment recommendation." A blue button asks "What is the total tumor volume?". Below this, a progress bar indicates "SUPPORTED" at 72%. The main content area displays the "Total segmented tumor volume is 118048.00 mm3 (118.05 cm3). Clinical implication: tumor burden tier tier is high risk. Overall segmented burden is high, which strengthens concern for clinically important intracranial mass effect." A source link "ctx:segmentation_metrics.total_tumor_volume_mm3" is provided. A red "HIGH RISK" banner states "Overall segmented burden supports a volumetric burden tier." and "Overall segmented burden is high, which strengthens concern for clinically important intracranial mass effect." Below the main content, there are several question prompts: "What is the total tumor volume?", "What is the enhancing tumor volume?", "What is the edema volume?", "Where is the tumor located?", "Which lobe is involved?", "How deep is the tumor from the cortical surface?", "What is the midline shift?", and "Is there ventricular compression?". A text input field for asking clinical questions is also present. The bottom of the interface shows a "Send" button and a status bar indicating "Connected" and "Locked to BraTS20_Validation_001".

Six Layers, One Direction

Data flows in one direction. No layer can alter numbers from the layer before it. 95% of code is shared across all three organs.



CARVIS Case Upload Screen



CARVIS Liver Slices Screen



The Output Is Verified, Not Trusted

`guardrails.py` reads every LLM output **line by line** and compares each measurement against the ground-truth context value.

Soft-Repair

Risky expressions are replaced via **soft-repair** rather than discarding the whole output. Example: *"chemotherapy"* → *"definitive management"*. If repair fails, the system falls back to the code-generated report.

Grounded Prompt

The prompt contains only a `FACTS_JSON` and a `REPORT_DESCRIPTOR_JSON`. The model cannot hallucinate numbers it was never given.

Hard Refusal

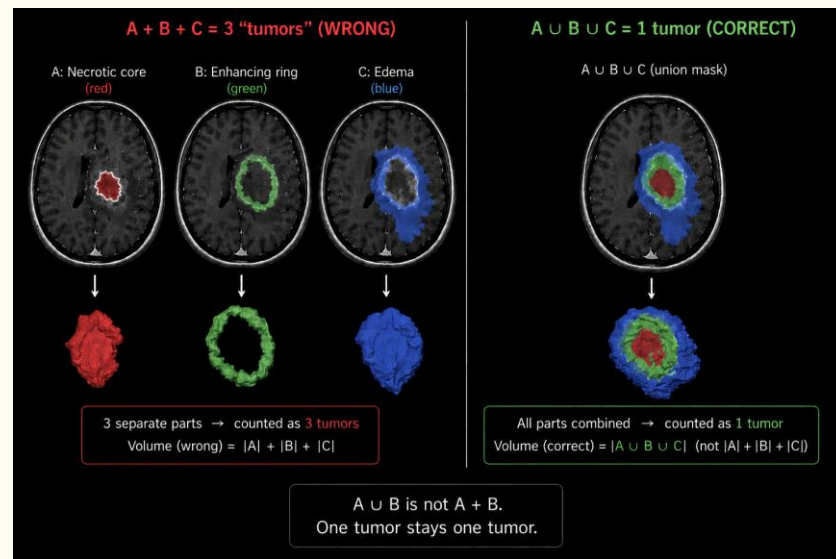
Guardrails rejects diagnosis, treatment, and prognosis questions outright. These intent classes are blocked by design : the system returns a refusal, not a fabricated answer.

☑ Most projects hope the model gets it right. **This one checks.**

$A \cup B$ Is Not $A + B$

A brain tumor has three overlapping components: necrotic core (BraTS label 1), edema (label 2), and enhancing ring (BraTS label 3 , **normalized to 1 before union** to prevent double-counting). A naive implementation counts each label separately and reports three lesions. That is clinically wrong.

- Components are merged into a **union mask** first, then passed to connected-component analysis.
- **Total volume** = voxel count of the union mask $\times$ voxel volume in mm^3 .
- **Diameter** = longest axis of the **3D bounding box** around the union mask ,not the largest single-label extent.
- Artifacts below the minimum voxel threshold are filtered before any measurement is made.



i The result reflects biological reality, not label arithmetic.

Every Answer Has a Source. Every Measurement Has a Range.

1

Contribution 03: Explainable QA

Each QA answer carries **evidence IDs** that resolve to specific field values in `evidence.json`. "Tumor diameter 38 mm" links to `ctx:max_diameter_mm = 38.2`. These surface as clickable tags in the UI. Explainable AI is a **coded feature**, not a slogan.

2

Contribution 04: Confidence Intervals

Derived from the model's real held-out **Dice score**: $\max(1 - \text{Dice}, 3\%)$.

- Brain Dice = 0.87 → **narrow interval**
- Liver & Lung Dice = 0.60–0.65 → **wider interval automatically**
- Confidence score hard-capped at **0.75 in code**

The system cannot claim certainty.



87%



65%



75%

Brain Dice

High segmentation accuracy, narrow confidence interval

Liver Dice

Moderate accuracy, wider confidence interval

Max Confidence Cap

Hard-coded ceiling, the system cannot overclaim

Intent Classification Without a Language Model

Eleven intent classes, classified by Jaccard token overlap against **six ontology groups: measurement, compartment, anatomy, reporting systems, reporting subtasks, and urgency**. No LLM involved.



Clinical Token Expansion

Before matching, clinical token expansion is applied, e.g. *"motor cortex"* expands to ["precentral", "eloquent", "m1"]. Structured reranking hints encode numerical targets (hemisphere, lobe, "<5 mm shift") to boost retrieval precision.



Per-Class Retrieval Policy

Each class has its own retrieval policy: a measurement question queries only the current case, a differential diagnosis question also pulls from the guideline corpus.



LLM Arbiter (Conditional)

The LLM arbiter activates under **three conditions**: top-2 class probability gap < 0.08 **or** a required slot is missing from the query **or** implicit urgency is detected. The expensive model is used only when the rule-based layer cannot resolve the question.

Search Architecture

Search combines BM25 and **all-MiniLM-L6-v2** sentence-transformers in parallel.

Conversation Memory

Conversation memory resolves follow-up questions like *"what about the edema?"* with a **1800-second session TTL** and no cross-case contamination.

CARVIS Follow Up Question Example

The screenshot displays the CARVIS Clinical AI Platform interface. On the left, a sidebar lists cases under the 'BRAIN' category, with 'BraTS20_Validation_001' selected. The main panel shows the 'CLINICAL Q&A' tab for this case. At the top, a 'Decision-support only. Not a medical diagnosis or treatment recommendation.' warning is present. Below it, a 'HIGH RISK' alert states: 'Overall segmented burden supports a volumetric burden tier. Overall segmented burden is high, which strengthens concern for clinically important intracranial mass effect.' A blue callout bubble asks 'What about the edema?'. The 'SUPPORTED' section shows a 72% confidence level and states: 'The largest burden component is edema at 97785.00 mm3 (97.78 cm3, 82.8% of total segmented volume)'. It lists four sources: 'ctx:segmentation_metrics.label_volumes_mm3.enhancing', 'ctx:segmentation_metrics.label_volumes_mm3.nonenhancing', 'ctx:segmentation_metrics.label_volumes_mm3.edema', and 'ctx:segmentation_metrics.total_tumor_volume_mm3'. A note below reads: 'For decision support only; clinician confirmation required.' Below this, a series of question buttons are displayed: 'What is the total tumor volume?', 'What is the enhancing tumor volume?', 'What is the edema volume?', 'Where is the tumor located?', 'Which lobe is involved?', 'How deep is the tumor from the cortical surface?', 'What is the midline shift?', and 'Is there ventricular compression?'. At the bottom, there is a text input field for asking a clinical question and a 'Send' button. The right sidebar shows the 'INSPECTOR' panel with case details and quick actions like 'View Report', 'Clinical Q&A', 'Open Imaging', and 'Print / PDF'. The status bar at the bottom indicates 'Connected' and 'Locked to BraTS20_Validation_001'.

CARVIS — Clinical AI Platform

CARVIS
CLINICAL AI

Search cases...

All | Brain | Lung | Liver

BRAIN 23

BraTS20_Validation_001

BraTS20_Validation_002

BraTS20_Validation_003

BraTS20_Validation_004

BraTS20_Validation_005

BraTS20_Validation_006

BraTS20_Validation_007

BraTS20_Validation_008

BraTS20_Validation_009

BraTS20_Validation_010

BraTS20_Validation_030

BraTS20_Validation_031

BraTS20_Validation_037

Upload New Case

Connected

BraTS20_Validation_001
FLAIR · T1 · T1ce · T2 · 118.85 cm³ burden · 3 lesions · Max Ø 89.9383 mm

Refresh

REPORT | **CLINICAL Q&A** | IMAGING

Decision-support only. Not a medical diagnosis or treatment recommendation.

HIGH RISK Overall segmented burden supports a volumetric burden tier.
Overall segmented burden is high, which strengthens concern for clinically important intracranial mass effect.

What about the edema?

SUPPORTED 72%

The largest burden component is edema at 97785.00 mm3 (97.78 cm3, 82.8% of total segmented volume).

4 SOURCES | ctx:segmentation_metrics.label_volumes_mm3.enhancing
ctx:segmentation_metrics.label_volumes_mm3.nonenhancing ctx:segmentation_metrics.label_volumes_mm3.edema
ctx:segmentation_metrics.total_tumor_volume_mm3

For decision support only; clinician confirmation required.

What is the total tumor volume? What is the enhancing tumor volume? What is the edema volume? Where is the tumor located?

Which lobe is involved? How deep is the tumor from the cortical surface? What is the midline shift? Is there ventricular compression?

Ask a clinical question... (Enter to send)

Locked to BraTS20_Validation_001

Send

INSPECTOR

CASE

ID BraTS20_Validation_001

Modality Brain MRI

QUICK ACTIONS

View Report

Clinical Q&A

Open Imaging

Print / PDF

STATUS

Report Ready

Q&A History 2 turns

Knowledge From Literature. Data Never Leaves the Room.

Grounded in Real Guidelines

Answers draw from actual guideline texts with every clinical threshold attributed:

- **RANO 2010, BT-RADS (AJNR 2017), NCCN CNS 2024**
- Midline shift ≥ 5 mm : Ropper 1986 NEJM
- Shift ≥ 7.5 mm herniation risk : Marmarou 1987
- Motor cortex ≤ 5 mm high risk : Duffau 2003
- Enhancing volume > 30 cm³ : Ellingson 2016

Privacy by Architecture

The entire system runs **offline**. Patient data never reaches a network.



KVKK Compliant



GDPR Compliant



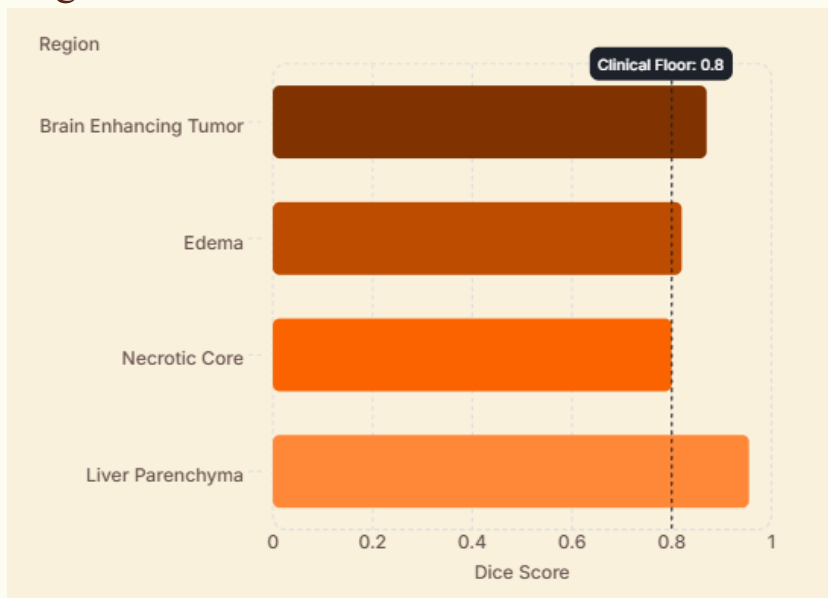
Deploy Today

✓ Compliant by architecture not by policy. It can go into a hospital today.

The System Measures Itself

Every run writes a **SHA256 fingerprint** of active code modules. The eval charts are generated automatically by 10 scripts across 5 modules. The system does not say it works , **it proves it.**

Segmentation Performance



System Integrity Metrics

Dice Floor: 0.80

Clinical minimum
across all segmentation
targets.

Hallucination Rate: 0.14

Measured generation
error rate under
evaluation.

Evidence Coverage: 0.94

Proportion of outputs
backed by cited
sources.

Confidence Cap: 75%

Maximum confidence
emitted ; prevents
overstatement.

What Was Actually Built

Three organs. One shared architecture. Fully local intent classification with 11 classes.

Core Components

01

API Layer

9 route files powering the backend interface.

02

Evaluation Suite

5 modules, 10 scripts : auto-generated eval charts.

03

Intent Classifier

11 classes, fully local : no external API calls.

04

Guideline Corpus

5 JSON files encoding clinical thresholds and sources.

05

Desktop App

Electron + React, Three.js mesh, vtk.js GPU volume rendering, PDF export.

06

Segmentation

3 organs : one shared architecture.

Models & Standards Under the Hood

Every component runs on the doctor's machine : no cloud services, no API calls.



Segmentation

nnU-Net v2 : BraTS gold standard, adapted per organ.



Narrative

Local LLM in **GGUF format** via llama.cpp; no API or cloud service.



Retrieval

all-MiniLM-L6-v2 + **BM25** in parallel; semantic and keyword scores blended.



Intent Classification

Pure rules and dictionaries; no model needed.

Anatomic Reference Standards

MNI152 T1 1mm : brain

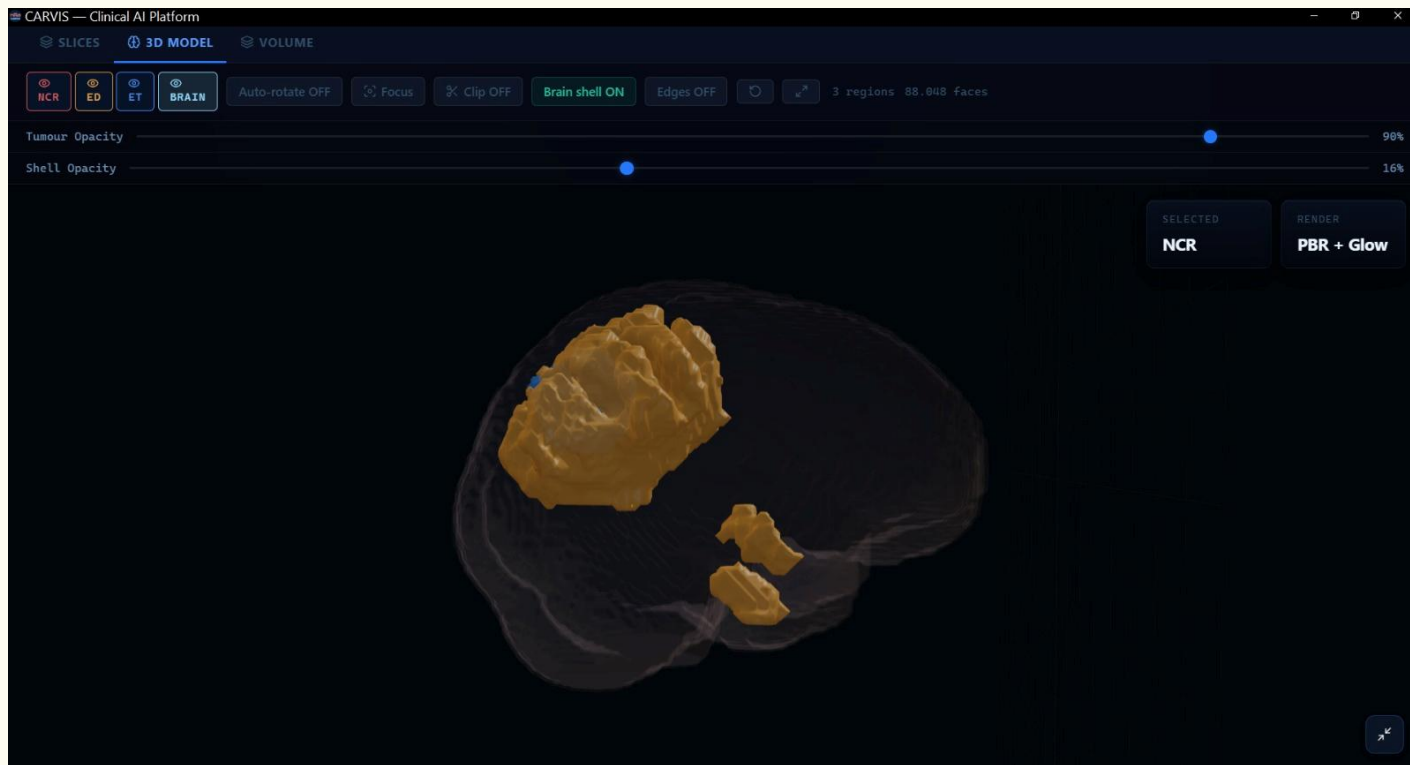
Harvard-Oxford Atlas : labels

Registration Pipeline (SimpleITK)

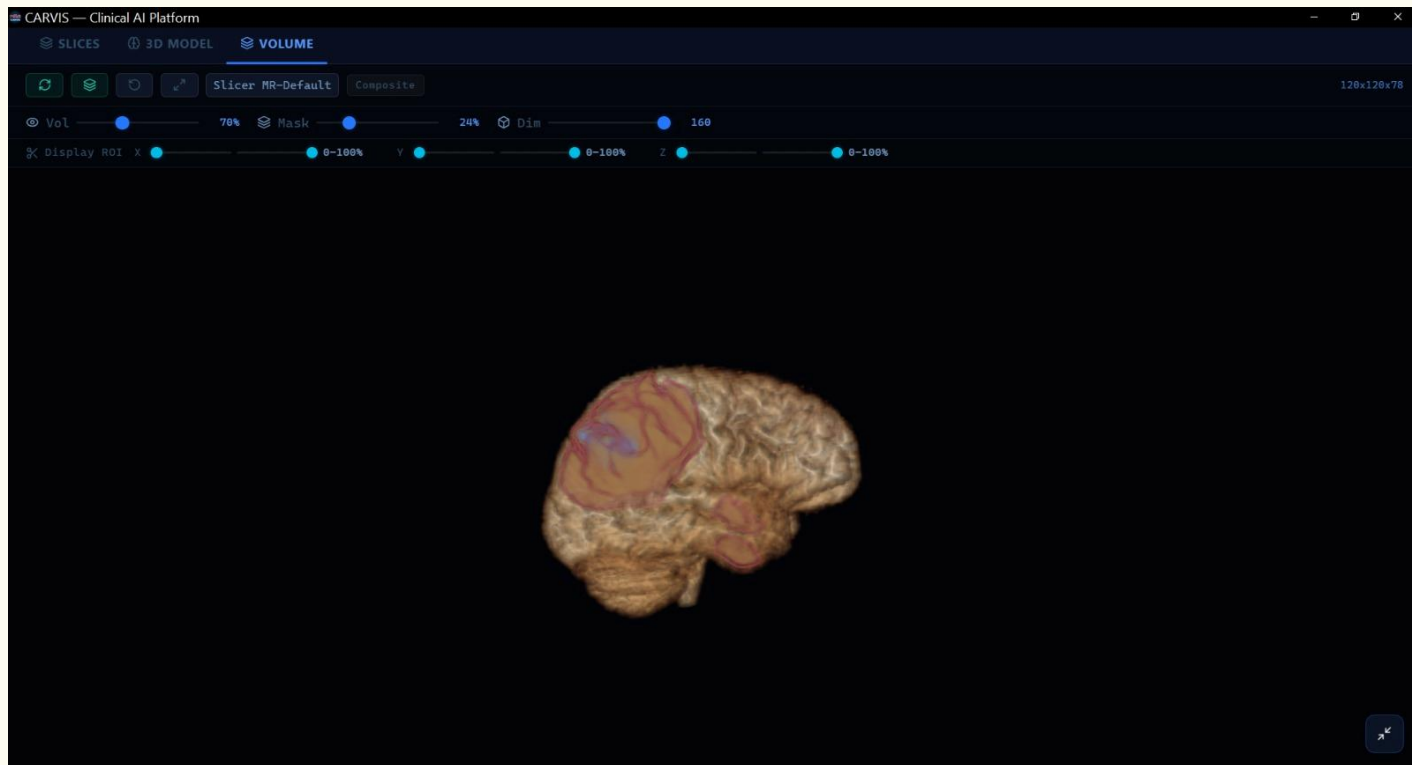
1. Rigid Alignment
2. Affine Refinement
3. Histogram Matching
4. Mattes Mutual Information

✓ The atlas, the models, the libraries , **everything runs on the doctor's machine.**

CARVIS Brain Mesh



CARVIS Brain Volume



CARVIS is Live

Two key resources are now publicly available : explore the platform and read the peer-reviewed research.

Website

www.carvisvr.com

Full platform documentation, evaluation metrics, and segmentation results across **BraTS 2020**, **MSD Lung**, and **Abdominal CT** datasets. Available now.

IEEE Publication

ieeexplore.ieee.org/document/11418476

"Transformative Technologies in Neurosurgery: a Systematic Literature Review"

✔ Peer-reviewed. Publicly accessible. Fully documented.

CARVIS Website QR



What We Built

Fluency

From the language model : natural, precise clinical language.

Accuracy

From the code : every threshold computed, not guessed.

Trust

From the validation layer : self-measuring, self-auditing, proven.

All three run offline, on your hardware, with your data staying in the room.

3

Organs

Brain, liver, and more : one shared segmentation architecture.

1

Architecture

Unified design across all targets and modules.

0

Network Calls

Fully offline. Patient data never leaves the room.

Real Medical Data, Inside a Headset

This is a VR medical visualization application built for the Meta Quest 3. It lets doctors explore real CT and MRI scans in an immersive 3D space.

Not slices on a screen. The actual volume, around you, at scale.

Real CT & MRI Scans

Actual patient data,
not synthetic models

Immersive 3D Space

Full volumetric
presence, not flat
slices

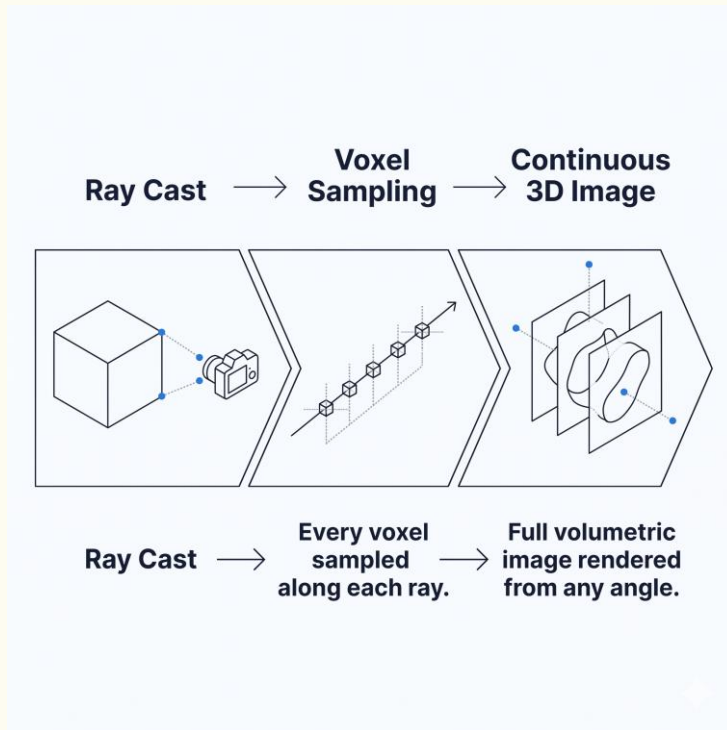
Meta Quest 3

Standalone, no PC required

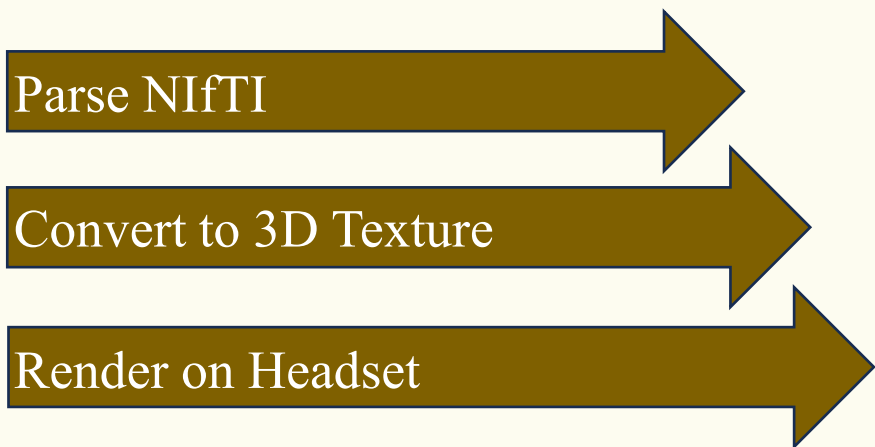
You See Inside the Body in Real Time

The app uses **GPU** ray-marching to render medical data as a true volume, not a stack of 2D slices. Every voxel is sampled along a ray cast from the camera.

The result is a continuous 3D image that you can look into from any angle. This runs entirely on the Quest hardware, in real time.



Real Patient Data, Loaded Directly



The app reads standard **NIfTI files**, the format hospitals and researchers actually use for CT and MRI scans.

The raw scan data is parsed and converted into a 3D texture that the GPU can sample directly during rendering.

- ✔ No preprocessing on a server. No conversion step on a PC. The headset handles it.

Grab It, Rotate It, Explore It



Grab the Volume

Pick up the scan with Quest controllers. The volume responds like a physical object in your hands.



Rotate Freely

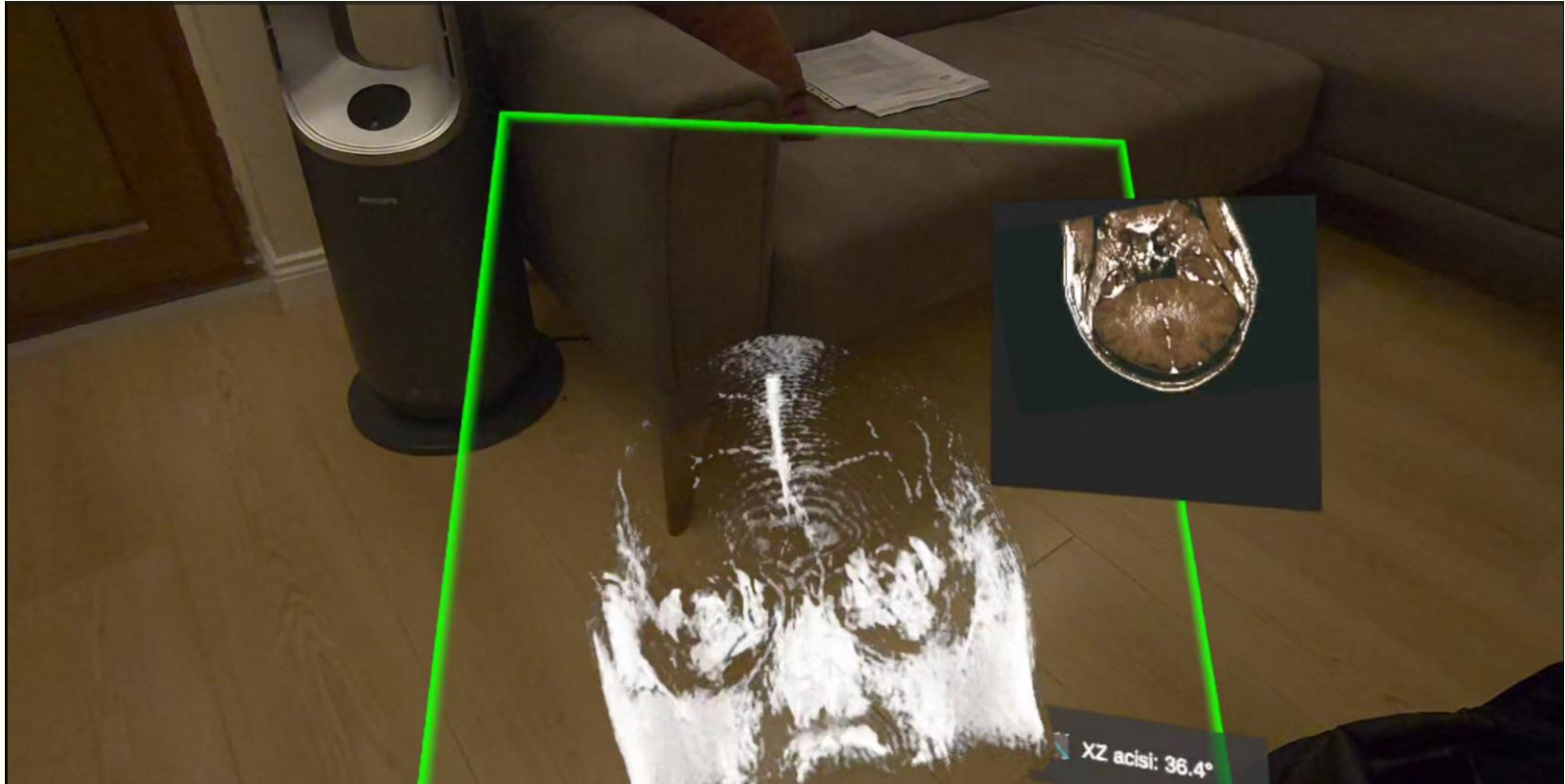
Spin the anatomy to any angle. No axis lock, no constraints , full spatial freedom.



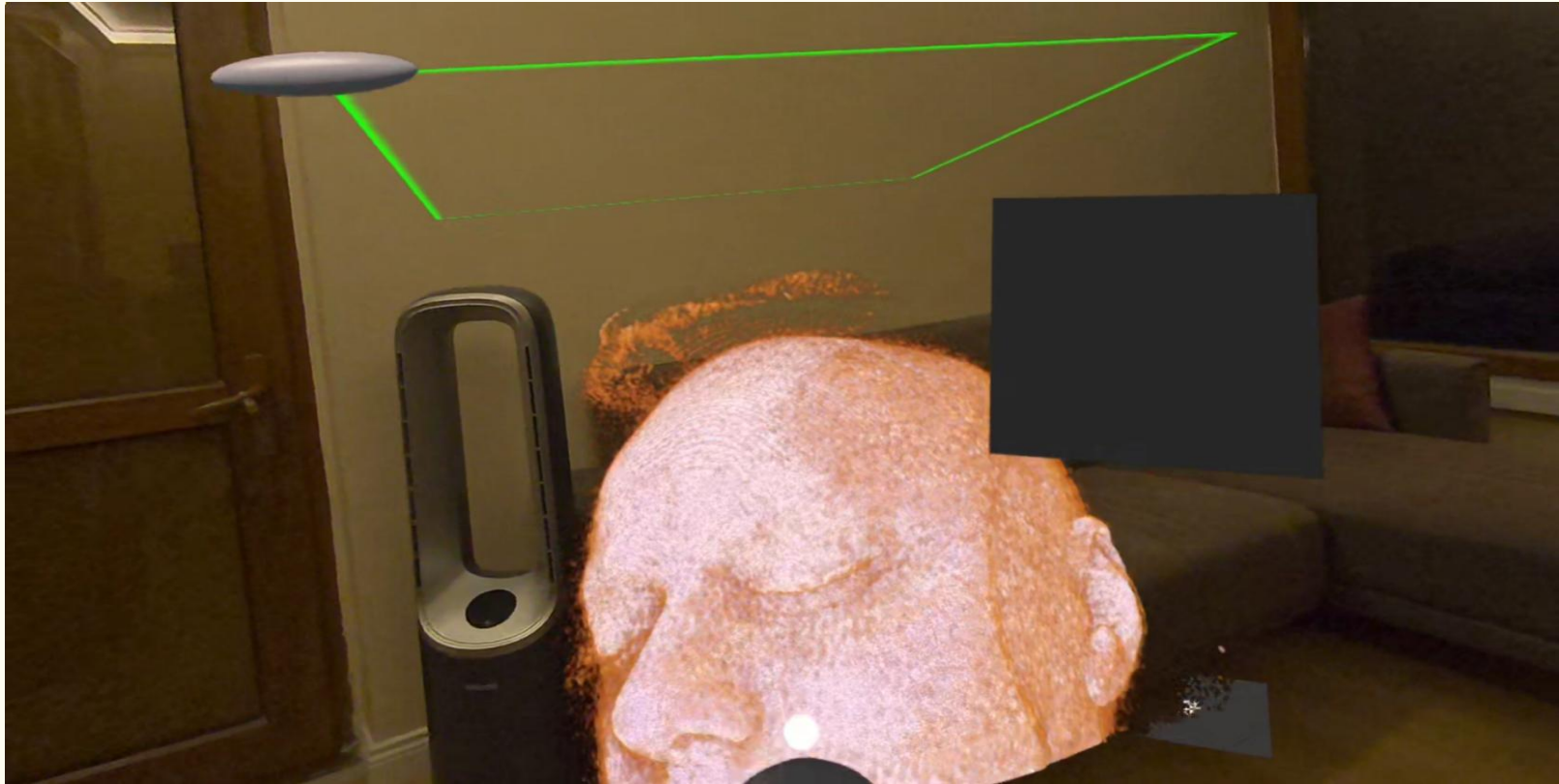
Unity New Input System

Clean, responsive controller input with no legacy overhead. Feels immediate and precise.

Real Time Volume Rendering



Real Time Volume Rendering

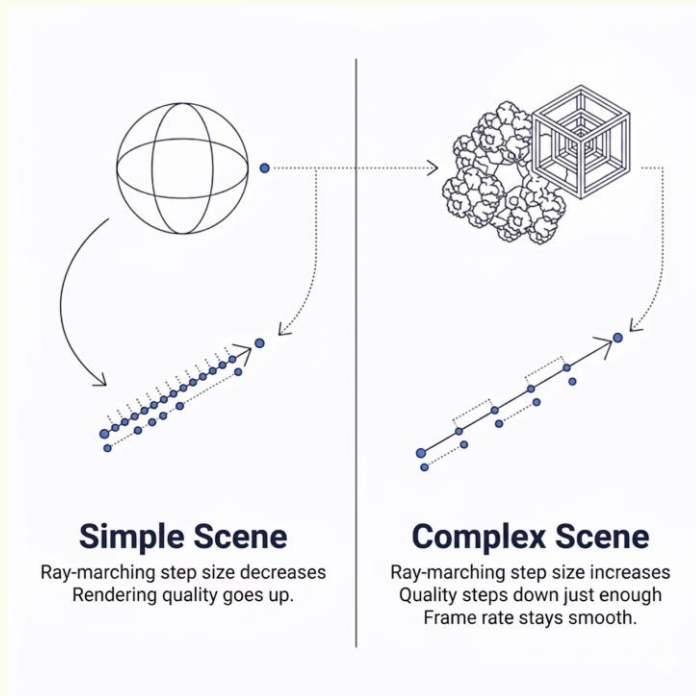


Built a Custom Optimizer for the Quest Hardware

Volume rendering is computationally expensive. Running it inside a VR headset at **high frame rates** is a real constraint.

To handle this we built **QuestVolumeOptimizer**, a dynamic sampling rate controller. It monitors frame time and adjusts the ray-marching step size on the fly.

- When the scene is simple, quality goes up
- When it gets complex, quality steps down just enough to keep framerate smooth
- No drops, no discomfort



Spatial Understanding Before the First Cut

The main use case is **surgical planning**. Before an operation a surgeon needs to understand the spatial relationship between a tumor, a vessel, a nerve.

The Old Way

Looking at 2D slices on a monitor. Mentally reconstructing depth and spatial relationships from flat images.

The CARVIS Way

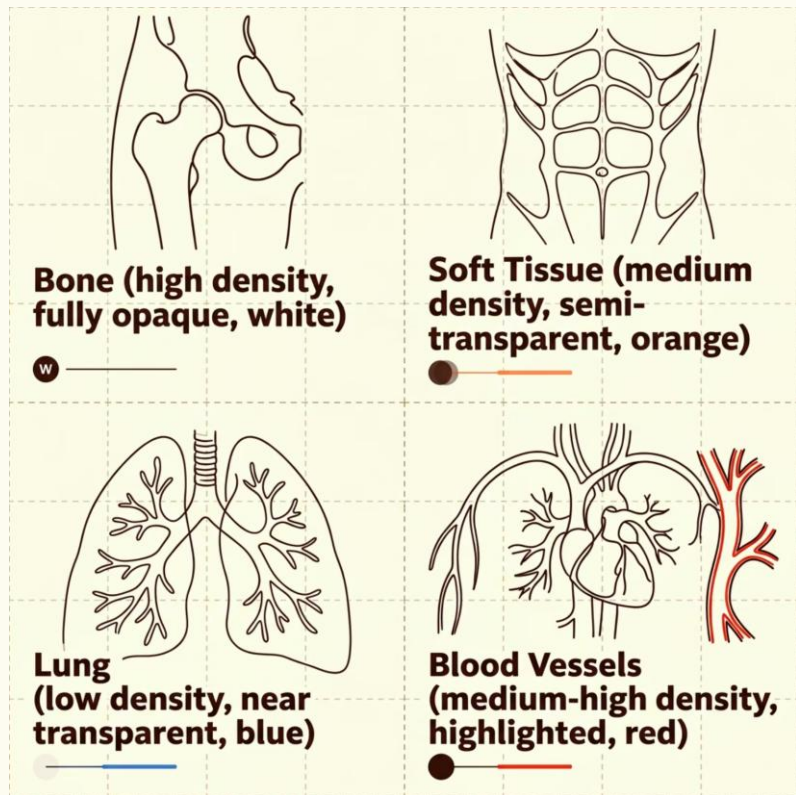
Standing inside a 3D volume of the actual patient anatomy. Examining the structure from any angle before ever making an incision.

Choose What You See

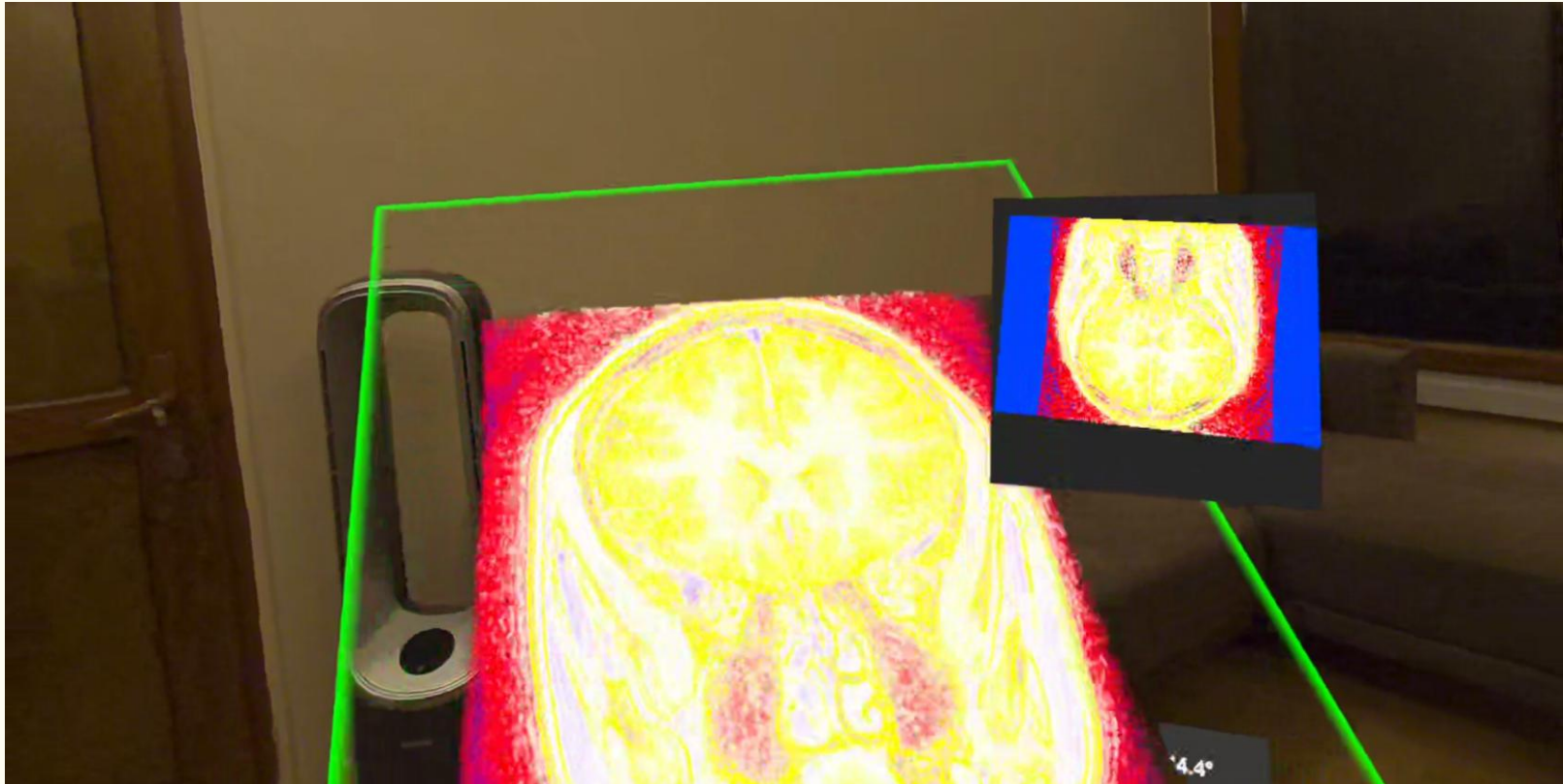
A **transfer function** maps each density value in the scan to a color and an opacity level. It decides which tissues appear solid and which fade to transparent.

Bone, soft tissue, lung, and blood vessels all have different density ranges. By choosing the mapping you choose what you are looking at.

The app ships with several presets so the doctor can switch focus across the same dataset instantly without reloading anything.



Real Time Volume Rendering



No Internet-No Server-No Cloud



Fully Offline

Zero internet required for rendering, data loading, or deployment.



Patient Data Stays On-Device

No upload step. No streaming. Data never leaves the headset.



No Remote Dependency

No dependency on a remote service. No backend to maintain or secure.



Just Works

Put the files on the headset and it works. That's the entire setup.

Native-Real Data-Inside the Headset

Everything you see is running natively on the headset, with real patient data. No PC tethered to it, no server behind it.

Just the scan, the GPU, and the doctor.

Native GPU Rendering

Ray-marching runs entirely on Quest hardware

Real NIfTI Data

Standard hospital CT and MRI formats

Fully Standalone

No PC, no server, no cloud

Thank You for Listening

We extend our sincere gratitude to those who made this work possible.

Advisor

We thank our advisor **Gül Tokdemir** for her invaluable guidance and support throughout this project.

Academic Support

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